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Influence of locking and passive soft switching on hydraulic circuit efficiency

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Abstract

Throttling and fluid compressibility losses are the major losses which occur during the valve transition in a switched mode hydraulic circuit. To minimize the throttling energy loss, a soft switch concept with lock and release mechanism was introduced by different researchers. In the literature, two different kinds of soft switches, one locking and another passive, were introduced in a hydraulic circuit. Thereafter, experimental validation was reported considering only a locking-type soft switch. But the influence of the passive soft switch on the performance of the hydraulic system was not reported. In this article, how the passive soft switch would affect the system efficiency has been investigated through bond graph model simulation. It has been shown both numerically and experimentally by other researchers that the use of a passive soft switch and a locking mechanism, depending upon configuration, can lead to more than 66.1% reduction in throttling energy loss. In this article, it has been further shown that an additional 3.25% throttling energy is saved when the passive soft switch is discarded from the system and only the locking mechanism is retained. The critical parameters of the locking soft switch have been optimized through multi-run simulation and, thereafter, a relationship between the critical parameters of the locking soft switch and the throttling energy loss of the valve has been obtained by using multiple regression analysis. Also, the variations of energy losses with respect to different valve orifice areas, duty cycles, and pulse width modulation frequencies are investigated through simulation.

Keywords

Soft switch, bond graphs, multiple regression analysis, throttling energy loss

1. Introduction

There has been a lot of research in recent years to minimize the throttling energy loss during the transition of the high-speed pulse-width-modulated (PWM) control valve in a hydraulic circuit. The rate of flow and average pressure can be maintained by the use of a PWM-controlled on/off valve.

Variable flow in a hydraulic circuit can be achieved by the use of a fixed displacement pump along with PWM control valve (on/off) instead of variable displacement pump.[1] Li et al. considered a hydro-mechanical analog of an electronic switched-mode power supply to produce an equivalent functional system of a variable displacement pump.[1] Their approach proved to be useful because of its success in switched mode converters in power electronics to obtain energy-efficient, compact, and economically better systems. However, on/off control valves possess less energy loss in fully opened and fully closed positions. The system which was proposed by Li et al. was more efficient than metering valve control system.[1] The above system was experimentally validated by Rannow et al.[2] This experimental validation confirmed that the variable flow could be obtained without the use of a variable displacement pump. In a power hydraulic circuit, various types of losses affect the performance of the system. In the study by Wu and Van de Ven,[3] different types of losses, i.e., throttling loss, compressibility loss, internal leakage loss, various frictional losses, etc., were studied in the case of a bidirectional valve. Generation and regeneration in both directions (forward and reverse) were considered, including the geometric design of a disk-style high-speed on/off valve for a switched-mode hydraulic system to obtain the highest efficiency.[3] The design of a unidirectional rotary valve with PWM control was reported by Tu et al.[4] The

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function of this valve was unique in the hydraulic circuit, where fluid was flowing through it as a power source for rotation. The design and the performance of the valve were validated experimentally by building a prototype.[4] In addition, Tu et al. analyzed the amount of loss through it and developed a mathematical model for simulation, which was also validated experimentally.[4] An approach to reduce the losses related to a high-speed on/off valve-controlled hydraulic system was presented by Rannow and Li.[5] A hydraulic system with a soft switch was modeled and simulated.[5] The simulated results ensured that the soft switching approach reduced a significant amount of throttling energy loss. The above approach also established that slower on/off valves were more effective.[5] The main challenge of this approach was to lock the soft switch and release it quickly with precise timing under loading conditions.[5] Van De Ven presented an idea on the lock and release mechanism of the soft switch,[6] where the pressure signal in the switched volume was used to trigger the release. This article focused on three different unique hydraulic circuits with locking and passive soft switch and analyzed the circuits with respect to minimum throttling energy loss. The most energy efficient hydraulic circuit among the three circuits was identified as containing both locking and passive soft switches.[6] Beckstrand and Van De Ven experimentally validated this efficient system,[7] considering only a locking soft switch.

In the study by Van De Ven,[6] both locking and passive soft switches were considered for simulation study in a hydraulic circuit. Later, the same concept was validated experimentally by Beckstrand and Van De Ven.[7] But that article did not identify the individual contributions toward energy saving of the two devices. Also, the earlier researchers have not specifically evaluated the influences of valve discharge coefficients and operating conditions on the energy saving. In this article, the influences of both locking and passive soft switches on high-speed valve controlled hydraulic circuits have been examined by using bond graph model simulation.[8–11] The bond graph models for two different hydrostatic transmission (HST) systems, one connected with the locking and passive soft switch (HST_{LKPS}) and another connected with the locking soft switch only, i.e., without passive soft switch (HST_{LK}), have been simulated. The parameters of the locking soft switch which can influence the performance of the system have been identified and optimized by using multi run simulation in SYMBOLS-Shakti software,[12] aiming toward minimum throttling energy loss. A relationship has been developed between the critical parameters of the locking soft switch and the throttling energy loss by multiple regression analysis using the multi run simulation data. Finally, the influences of different valve orifice areas, duty cycles and operating frequencies of the pulse width modulation (PWM) have been studied through model simulation.

2. Operation of the high speed valve controlled switched mode hydraulic circuit

The hydraulic circuits shown in Figure 1(a) and (b) are the HST systems in which a constant flow (Q_{supl}) is supplied by a hydraulic pump to the tank or load valve. The flow is passed through the tank or the load valves as per the requirement. When the valves start to switch from the tank valve to load valve or vice versa, both the locking soft switch (connected between hydraulic pump and the tank) and the passive soft switch (connected with hydraulic pump) start to work. The main function of the soft switch is to absorb the normally throttled fluid into a small chamber when the valve is partially closed, and release the absorbed fluid when the valve is fully opened. This soft switch mechanism helps to minimize the throttling loss during the valve transition in the hydraulic circuit. The detail operation of the soft switch has been described by Van de Ven.[6] The components of the locking soft switch are: one piston, piston cylinder and one helical compression spring. Referring to Figure 1, two check valves are connected at the rear end of the locking soft switch. One is used to supply the fluid to the locking soft switch when the rear end pressure of the locking soft switch is less than the tank pressure and another one is used for the safety purpose. If the pressure developed inside the locking soft switch is high enough, it will release the pressure by flowing fluid to the high pressure rail. The function of the passive soft switch is also the same as that of the locking soft switch, but it is connected with only pump delivery line. The operation of the HST_{LK} system (see Figure 1(b)) is just similar to the HST_{LKPS} system (i.e., Figure 1(a)), except the passive soft switch which is removed from the circuit in Figure 1(a). An enlarged view of the assembly of the locking soft switch is shown in Figure 1(c). The locking soft switch consists of one cylindrical-piston assembly and one pre-compressed helical spring of stiffness K_{ssl}.

Both the systems are controlled by PWM given to the valve spool. The sequence of operation of the valve is described below (see Figure 2).

After a certain period of operation through the tank valve, the tank valve starts to close from point 1 (see Figure 2) and is fully closed at point 2. The total time duration for closing the valve is t_1. During the closing of tank valve, the switched volume pressure is increased and the piston of the soft switch starts to travel towards the rear end of the soft switch. After complete closure of the tank valve, the load valve starts to open from point 2 and it is fully opened at point 3. The total time duration to fully open is t_2. The load valve continues to retain its full open condition between the point 3 and 4 (see Figure 2). Thereafter, the load valve starts to close from point 4 and is closed completely at point 5. The total transition time elapsed for closing the load valve is t_3. Proceeding further,

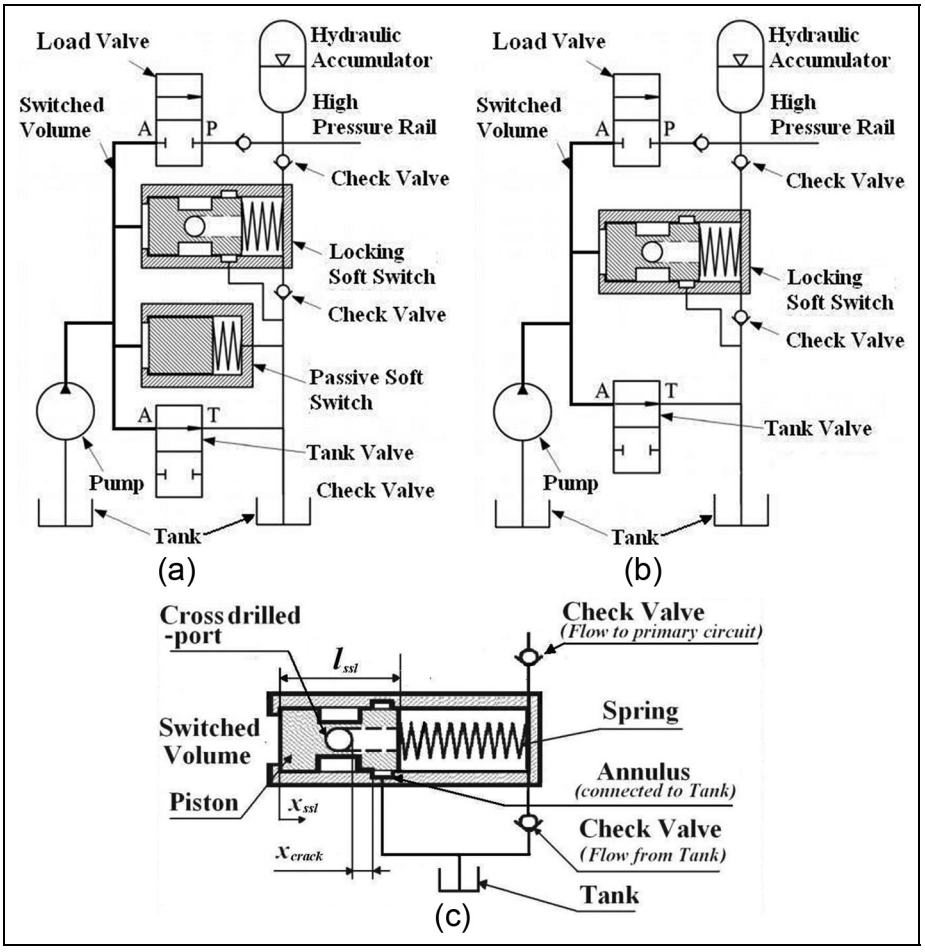

Figure 1. (a) Hydraulic circuit with locking and passive soft switch (HST_{LKPS}),[6] (b) hydraulic circuit without passive soft switch (HST_{LK}),[7] and (c) enlarged view of the locking soft switch.

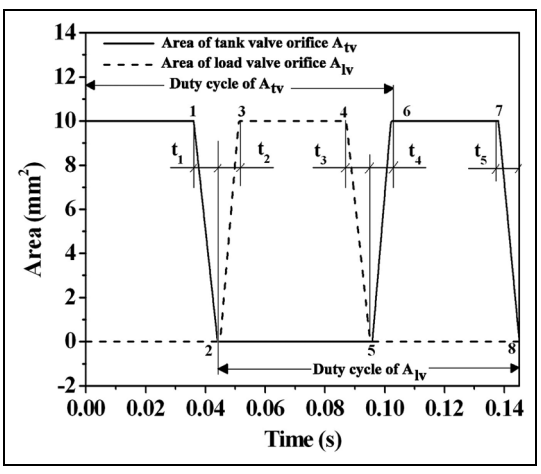

Figure 2. Orifice area of the tank and the load valve as a function of time for one PWM cycle.

the tank valve orifice start to open at point 5. The pressure surge in the switched volume is maximum at point 5. Due to the high pressure, the port of the locking soft switch piston coincides with the internal passage of the soft switch wall and permits the working fluid to flow from rear end to the tank. As a result, the fluid tapped at the rear end is released and unlocking of the piston is possible. Now, the piston of the soft switch returns back to its locked position due to the force equilibrium, being ready for the next cycle. The time taken to reach the fully open condition is t_4. Therefore, the total transition processes in one PWM are four. The same sequences are continued for further cycles.

3. Modeling of the hydraulic system

3.1 Hydraulic circuit with locking and passive soft switch

A portrayal representation of any physical dynamic system can be done by using bond graphs.[8–11] It involves transforming the system into a dynamic simulation model in source code form and it is popularly used by different researchers for analyzing systems having the interaction of

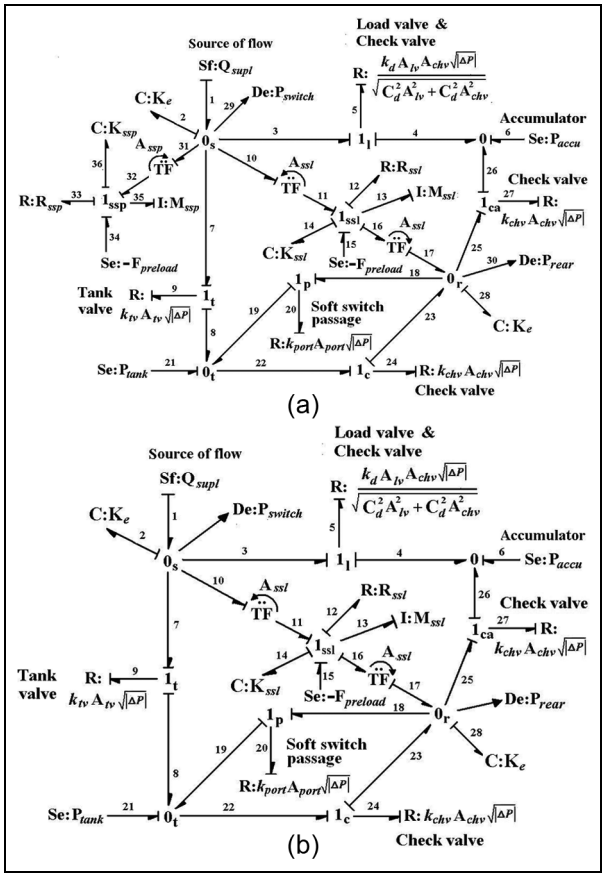

Figure 3. Bond graph models: (a) hydraulic circuit with locking and passive soft switch (HST_{LKPS}); (b) hydraulic circuit without passive soft switch (HST_{LK}).

multi energy domains.[13,14] The bond graph models for both the systems are shown in Figure 3. The following assumptions are made in the model development.

(1) The inertia of the working fluid is negligible.
(2) The dynamics of the valves are not addressed.
(3) A constant flow source is always maintained at the inlet of the valve.
(4) The soft switch ports and valve orifices are considered as rectangular.
(5) Fluid properties are assumed to be unaffected due to the variation of temperature.

In Figure 3, Sf_1 represents the constant flow supply Q_{supl} from fixed displacement hydraulic pump which passes through the tank valve/load valve as per the requirement. In the bond graph model, these two valves (i.e., tank valve and load valve) are modeled by R elements at 1_t and 1_1 junctions, i.e., R_9 and R_5. Note that R_5 represents the resistance of flow through the load valve and check valve together because both valves are connected in series in the hydraulic circuit (see Figure 1). The switched pressure (P_{switch}) of the flowing fluid in the circuits is measured by a detector (De_{29}). The effective fluid bulk modulus (K_e) of the switched volume is presented by C element at 0_s junction. The transformer element (TF) connected between 0_s and 1_{ssl}, converts the hydraulic energy into mechanical energy and builds an unbalanced force into the piston of the soft switch in case of sudden increase of P_{switch}. The dynamics of the locking soft switch is modeled at 1_{ssl} junction by appending the preload of the spring ($F_{preload}$), the stiffness of the spring (K_{ssl}), the mass of the piston (M_{ssl}) and the resistance (R_{ssl}) between the soft switch piston and the wall of the soft switch (Figure 3(a) and 3(b)). Another TF connected between 1_{ssl} and 0_r, converts the mechanical energy into the hydraulic energy. The rear end pressure (P_{rear}) of the locking soft switch is measured at 0_r junction by using effort detector De_{30}. The effective bulk modulus (K_e) of the flowing fluid at the rear end is represented by a C element (C_{28}). The check valve (tank to the rear end of the soft switch) and the resistance of flow through the internal passage of the soft switch are modeled by R elements, i.e., R_{24} and R_{20} at 1_c and 1_p junction, respectively. In the same way, the check valve (rear end of the soft switch to high pressure line) which is used for safety purpose is modeled by another R element at 1_{ca} junction. The pressures, i.e., P_{tank} and P_{accu} are the tank and the accumulator pressure, respectively and are represented by the Se elements.

The dynamics of the passive soft switch (see Figure 3(a)) is modeled at 1_{ssp} junction by appending to it the preload of the spring ($F_{preload}$), the stiffness of the spring (K_{ssp}), the mass of the piston of the soft switch (M_{ssp}) and the resistance (R_{ssp}) which are represented by Se, C, I and R elements, respectively.

The general nonlinear flow equation for all check valves, tank valve, load valve, and port of the locking soft switch can be expressed as:

$$Q = k_c A_o \sqrt{|\Delta P|}\,\mathrm{sign}\,(\Delta P), \tag{1}$$

where $k_c = C_d\sqrt{2/\rho}$ is the flow constant whose numerical value depends on the coefficient of discharge (C_d) and mass density (ρ) of the flowing fluid. The A_o is the area of the orifice of the passage, while ΔP is the pressure difference across the valve port.

The flow through the tank valve and the load-check valve (connected in series) are given by:

$$R_9 : Sf_9 = k_{tv} A_{tv} \sqrt{|(P_{switch} - P_{tank})|}\,\mathrm{sign}(P_{switch} - P_{tank}), \tag{2}$$

$$R_5 : Sf_5 = \frac{k_d A_{lv} A_{check}}{\sqrt{(C_d A_{lv})^2 + (C_d A_{check})^2}} \sqrt{|(P_{switch} - P_{accu})|}\,\mathrm{sign}(P_{switch} - P_{accu}). \tag{3}$$

In the above equations, i.e., Equations (2) and (3), k_{tv} and k_d are the flow constants of the tank valve and equivalent flow constant of the load-check combination, respectively. Similarly, A_{tv}, A_{lv}, and A_{check} are the port opening areas of the tank, load, and check valves, respectively.

The piston velocities and of the locking soft switch ($\dot{x}_{ssl} = v_{ssl}$) and the passive soft switch ($\dot{x}_{ssp} = v_{ssp}$) are obtained by solving the ordinary differential equations (ODEs) obtained from the bond graph models:

$$\dot{v}_{ssl} = \frac{(P_{switch} - P_{rear})A_{ssl} - F_{preload} - K_{ssl}x_{ssl} - R_{ssl}v_{ssl}}{M_{ssl}}, \tag{4}$$

$$\dot{v}_{ssp} = \frac{P_{switch}A_{ssp} - F_{preload} - K_{ssp}x_{ssp} - R_{ssp}v_{ssp}}{M_{ssp}}, \tag{5}$$

where A_{ssl}, x_{ssl}, v_{ssl} and A_{ssp}, x_{ssp}, v_{ssp} are the base area of the piston, linear axial displacement, and the linear velocity of the piston of the locking and the passive soft switch, respectively. The viscous resistances for the locking soft switch (R_{ssl}) and the passive soft switch (R_{ssp}), are expressed as $R_{ssl} = \frac{2\pi\mu r_{ssl}l_{ssl}}{C_{ss}}$ and $R_{ssp} = \frac{2\pi\mu r_{ssp}l_{ssp}}{C_{ss}}$, respectively. In the above expression, r_{ssl}, l_{ssl} and r_{ssp}, l_{ssp} (see Figure 1) are the radius and length of the piston of the locking soft switch and the passive soft switch, respectively. In this study, the materials considered for the spring and the piston are oil hardened tamper spring steel and steel, respectively. The C_{ss} and μ are the bore clearance of the piston and the dynamic viscosity of the working fluid, respectively. Note that Equation (5) and the related terms of the passive soft switch are not valid for Figure 3(b).

The switched volume pressure (P_{switch}) as per Figure 3(a) is expressed as follows:

$$P_{switch} = \int \left\{ \frac{K_e(P_{switch})}{(V_{switch} + A_{ssl}x_{ssl} + A_{ssp}x_{ssp})} \left(Q_{pump} - Q_{tv} - Q_{lv} - A_{ssl}v_{ssl} - A_{ssp}v_{ssp}\right) \right\} dt, \tag{6}$$

where V_{switch} is the switched volume and Q_{tv} and Q_{lv} are the rates of flow through tank valve and load valve, respectively. When the passive soft switch is absent in the hydraulic circuit (see Figure 3(b)), related terms of the passive soft switch are removed from Equation (6).

The fluid bulk modulus refers to the resistance of a liquid to compression and is the reciprocal of compressibility. The bulk modulus is an essential and inbuilt property of a liquid whose numerical value depends on various factors: operating conditions; the amount of entrained air/gas present; and compression process.[15,16]

The effective bulk modulus (K_e) of the working fluid depends on the pressure and is expressed as follows:

$$K_e(P) = K\left[\frac{\left(P/P_0\right)^{\left(1/\gamma\right)} \ell^{\frac{(P-P_0)}{K}} + R}{\frac{RK}{\gamma P} + \left(P/P_0\right)^{\left(1/\gamma\right)} \ell^{\frac{(P-P_0)}{K}}}\right], \tag{7}$$

where K is the bulk modulus of the air-free flowing fluid, P_0 is atmospheric pressure, and γ (=1.4) and R (=0.02) are ratio of specific heat of air and volume fraction of air entrained in the oil, respectively.

The pressure at the rear end (P_{rear}) of the locking soft switch can be expressed as follows:

$$P_{rear} = \int \left\{ \frac{K_e(P_{rear})}{V_{ssl_clear} + A_{ssl}(x_{ssl_max} - x_{ssl})} \left(A_{ssl}v_{ssl} + Q_{t_chv} - Q_{chv_accu} - Q_{ssl_port}\right) \right\} dt, \tag{8}$$

In Equation (8), x_{ssl_max} and V_{ssl_clear} are the maximum possible travel length and the clearance volume at the rear end of the locking soft switch, respectively. The flows Q_{t_chv} (tank to the rear end of locking soft switch) and $Q_{chv-accu}$ (rear end of locking soft switch to the accumulator) are passed through two check valves. The flow rate through the internal passage of the locking soft switch to the tank is denoted by Q_{ssl_port}. The above flow rates can be expressed as follows:

$$Q_{t_chv} = \begin{cases} 0, & \text{if } P_{crack} > (P_{tank} - P_{rear}) \\ k_{check}A_{check}\sqrt{|(P_{tank} - P_{rear} - P_{crack})|}\,\text{sign}(P_{tank} - P_{rear} - P_{crack}), & \text{if } P_{crack} \leqslant (P_{tank} - P_{rear}) \end{cases}, \tag{9}$$

$$Q_{chv_accu} = \begin{cases} 0, & \text{if } P_{crack} > (P_{rear} - P_{accu}) \\ k_{check}A_{check}\sqrt{|(P_{rear} - P_{accu} - P_{crack})|}\,\text{sign}(P_{rear} - P_{accu} - P_{crack}), & \text{if } P_{crack} \geqslant (P_{rear} - P_{accu}) \end{cases}, \tag{10}$$

$$Q_{ssl_port} = k_{port}A_{ssl_port}\sqrt{|(P_{rear} - P_{tank})|}\,\text{sign}(P_{rear} - P_{tank}). \tag{11}$$

In Equations (9)–(11), $k_{check}(= C_d\sqrt{2/\rho})$ is the flow constant of the check valve and is a function of the coefficient of discharge C_d and the mass density ρ of the working fluid. The opening area of the orifice is denoted by A_{check}. Similarly, k_{port} is the flow constant corresponding to the fluid flow from the rear end of the locking soft switch to

the tank when the annular portion coincides with the internal port area $A_{\text{ssl_port}}$. The check valve opening areas are assumed to be instantaneous, whereas the internal passage opening area $A_{\text{ssl_port}}$ of the locking soft switch is conditional and expressed as follows:

$$A_{\text{ssl_port}} = \begin{cases} 0, & \text{if } (x_{\text{ssl}} < x_{\text{crack}}) \\ 2\pi r_{\text{ssl}}(x_{\text{ssl}} - x_{\text{crack}}), & \text{if } \left(x_{\text{crack}} \leqslant x_{\text{ssl}} < x_{\text{crack}} + l_{\text{ss_port}}\right), \\ 2\pi r_{\text{ssl}}\, l_{\text{ssl_port}}, & \text{if } \left(x_{\text{ssl}} \geqslant x_{\text{crack}} + l_{\text{ss_port}}\right) \end{cases} \tag{12}$$

where $l_{\text{ssl_port}}$ and x_{crack} are the length of the internal passage of the locking soft switch and the crack distance travelled by the locking soft switch piston to open the passage to tank for pressurized fluid (accomplishing unlocking), respectively.

From the bond graph model (given in Figure 3(a)), the state equations of the system are derived as follows:

$$\begin{aligned} &Q_{\text{supl}} - A_{\text{ssl}}v_{\text{ssl}} - A_{\text{ssp}}v_{\text{ssp}} \\ &- k_{\text{tv}}A_{\text{tv}}\sqrt{|P_{\text{switch}} - P_{\text{tank}}|}\,\text{sign}(P_{\text{switch}} - P_{\text{tank}}) \\ &- \frac{k_d A_{\text{lv}} A_{\text{check}}\sqrt{|P_{\text{switch}} - P_{\text{accu}}|}\,\text{sign}(P_{\text{switch}} - P_{\text{accu}})}{C^2{}_d A^2{}_{\text{lv}} + C^2{}_d A^2{}_{\text{check}}} \\ &- \frac{\dot{P}_{\text{switch}}}{K_e} = 0 \end{aligned} \tag{13}$$

$$\begin{aligned} &A_{\text{ssl}}v_{\text{ssl}} + k_{\text{check}}A_{\text{check}}\sqrt{|P_{\text{tank}} - P_{\text{rear}}|}\,\text{sign}(P_{\text{tank}} - P_{\text{rear}}) \\ &- k_{\text{port}}A_{\text{ssl_port}}\sqrt{|P_{\text{rear}} - P_{\text{tank}}|}\,\text{sign}(P_{\text{rear}} - P_{\text{tank}}) \\ &- k_{\text{check}}A_{\text{check}}\sqrt{|P_{\text{rear}} - P_{\text{accu}}|}\,\text{sign}(P_{\text{rear}} - P_{\text{accu}}) \\ &- \frac{\dot{P}_{\text{rear}}}{K_e} = 0 \end{aligned} \tag{14}$$

$$\begin{aligned} &(P_{\text{switch}} - P_{\text{rear}})A_{\text{ssl}} - F_{\text{preload}} - K_{\text{ssl}}x_{\text{ssl}} \\ &- R_{\text{ssl}}v_{\text{ssl}} - M_{\text{ssl}}\dot{v}_{\text{ssl}} = 0, \end{aligned} \tag{15}$$

$$\begin{aligned} &P_{\text{switch}}A_{\text{ssp}} - F_{\text{preload}} - K_{\text{ssp}}x_{\text{ssp}} - R_{\text{ssp}}v_{\text{ssp}} \\ &- M_{\text{ssp}}\dot{v}_{\text{ssp}} = 0. \end{aligned} \tag{16}$$

The state equations for the other system (i.e., HST_{LK}) are of the same structure as Equations (13)–(16) after deleting the passive soft switch terms (i.e., A_{ssp}, x_{ssp}, v_{ssp}, R_{ssp}, K_{ssp}) from those equations. The equations are directly obtained from the corresponding bond graph model shown in Figure 3(b).

4. Results and discussion

The system equations, i.e., Equations (13)–(16), derived from the bond graph models (Figure 3(a) and (b)) have been solved numerically by the use of Symbols-Shakti software.[12] The parameter values assigned for simulation of the bond graph models are given in Table 1.

Parameter optimization is a method used to achieve the optimum solution or unconstrained maxima or minima of continuous and differentiable functions. Optimization for

Table 1. Parameters for the simulation.

Description	Value
Flow rate of pump, Q_{supl}	4 L min^{-1}
Tank pressure, P_{tank}	1.013×10^5 N m^{-2}
Accumulator pressure, P_{accu}	20.7×10^6 N m^{-2}
Atmospheric pressure, P_0	1.013×10^5 N m^{-2}
Mass density of working fluid, ρ	876 kg m^{-3}
Bulk modulus of working fluid, K	1.9×10^9 N m^{-2}
Dynamic viscosity of working fluid, μ	0.59 N s m^{-2}
Delay time transition for both valves	2 ms
Area of tank valve, A_{tv}	10×10^{-6} m^2
Area of load valve, A_{lv}	10×10^{-6} m^2
Area of check valve, A_{check}	8×10^{-6} m^2
Load valve flow constant, k_{lv}	0.02756 kg$^{-0.5}$ m$^{1.5}$
Tank valve flow constant, k_{tv}	0.0304 kg$^{-0.5}$ m$^{1.5}$
Check valve flow constant, k_{check}	0.044×10^{-6} kg$^{-0.5}$ m$^{1.5}$
Cracking pressure of the check valve, P_{crack}	20.7×10^3 N m^{-2}
Piston radius, $r_{\text{ssl}} = r_{\text{ssp}}$	4.5×10^{-3} m
Length of the piston of the soft switch, $l_{\text{ssl}} = l_{\text{ssp}}$	11×10^{-3} m
Stiffness of the passive soft switch spring, K_{ssp}	1×10^7 N m^{-1}
Clearance volume at the rear end of the soft switch, $V_{\text{ssl_clear}}$	1×10^{-6} m^3
Switched volume, V_{switch}	15×10^{-6} m^3
Radial clearance of the piston, C_{ss}	10×10^{-6} m

Table 2. Regression data.

Sl. No.	x_{crack}	K_{ssl}	$F_{preload}$	x_{ssl_max}	$E_{throttloss}$
1	0.002	100000	10	0.01	3.60984
2	0.002047	83147.4	10.3436	0.010296	3.10117
3	0.002096	69134.8	10.699	0.0106	2.82578
4	0.002145	57483.8	11.0666	0.010914	2.59704
5	0.002196	47796.3	11.4468	0.011237	2.39945
6	0.002248	39741.3	11.8401	0.011569	2.22952
7	0.002302	33043.9	12.2469	0.011911	2.0784
8	0.002356	27475.1	12.6677	0.012264	1.94949
9	0.002412	22844.8	13.103	0.012627	1.83878
10	0.002469	18994.9	13.5532	0.013	1.74482
11	0.002527	15793.7	14.0189	0.013385	1.66596
12	0.002587	13132.1	14.5005	0.013781	1.60122
13	0.002649	10919	14.9987	0.014188	1.55017
14	0.002711	9078.84	15.5141	0.014608	1.51269
15	0.002775	7548.82	16.0471	0.01504	1.48991
16	**0.002841**	**6276.64**	**16.5985**	**0.015485**	**1.48319**
17	0.002908	5218.86	17.1688	0.015943	1.49572
18	0.002977	4339.35	17.7587	0.016415	1.53653
19	0.003048	3608.05	18.3689	0.0169	1.61584
20	0.00312	3000	19	0.0174	1.81754

hydraulic systems is an important aspect and has been studied by various researchers.[17–20] Optimization can be done with various objectives and also using various methods. Off-road vehicles used in industry consume a lot of energy and are generally powered by non-green sources such as diesel and other fossil fuels. This contributes to pollution as well as global warming. This article deals with parameter optimization for energy saving to reduce environmental footprint. In this article, two bond graph models of the hydraulic circuits, HST_{LKPS} (Figure 3(a)) and HST_{LK} (Figure 3(b)) have been simulated with optimized critical parameters of the locking soft switch, obtained by multiple run simulation. The critical parameters, which are taken into consideration for optimization with respect to minimum throttling energy losses, are spring stiffness (K_{ssl}), preloading force of the spring ($F_{preload}$), crack length of the piston travel (x_{crack}), and maximum possible travel of the piston (x_{ssl_max}). An analytical relationship has been developed to correlate the critical parameters and throttling energy loss by using multiple regression analysis, as discussed in Section 4.1.

4.1. Multiple regression analysis

Multiple regression analysis helps to build a relationship among the parameters whose numerical values are critical during simulation. In this study, the identified independent critical parameters of the locking soft switch (see Figure 1(c)) are crack length of the piston travel (x_{crack}), stiffness of the spring (K_{ssl}), preload of the spring ($F_{preload}$), maximum possible travel of the piston (x_{ssl_max}), and the dependent variable is throttling energy loss ($E_{throttloss}$). The relationship between those variables is assumed to be as follows:

$$E_{throttloss} = a_0 + a_1 x_{crack} + a_2 K_{ssl} + a_3 F_{preload} + a_4 x_{ssl_max}, \tag{17}$$

where a_i (i = 0–4) are constants.

The regression data obtained through multi-run simulation are given in Table 2. The constants of the regression equation (Equation (17)) are solved by using Origin software. The equation is given as follows:

$$\begin{aligned} E_{throttloss} = & -209 + 1.297 \times 10^6 x_{crack} \\ & + 3.85 \times 10^{-5} K_{ssl} + 196.53 F_{preload} \\ & - 4.35 \times 10^5 x_{ssl_max}. \end{aligned} \tag{18}$$

The value of regression coefficient (R^2) is found to be 0.996. Therefore, this model is useful from the accuracy viewpoint.

In Table 2, the row (Sl. No. 16) corresponding to minimum throttling energy loss ($E_{throttloss}$) is highlighted in bold. The locking soft switch parameter values (i.e., x_{crack}, K_{ssl}, $F_{preload}$, and x_{ssl_max}), as given in Sl. No. 16, have been assigned in the model simulation (discussed in Section 3). During multi run simulation, spring stiffness is varied from 1×10^5 N/m to 3×10^3 N/m. With the decrease of the spring stiffness value, x_{crack} is increased and throttling energy loss is reduced or vice-versa. If the spring stiffness is more than 1×10^5 N/m, the x_{crack} value is less than 2 mm, which is difficult to fabricate the soft switch. On the contrary, when the spring stiffness is less than 3×10^3 N/m, residual error is developed during the

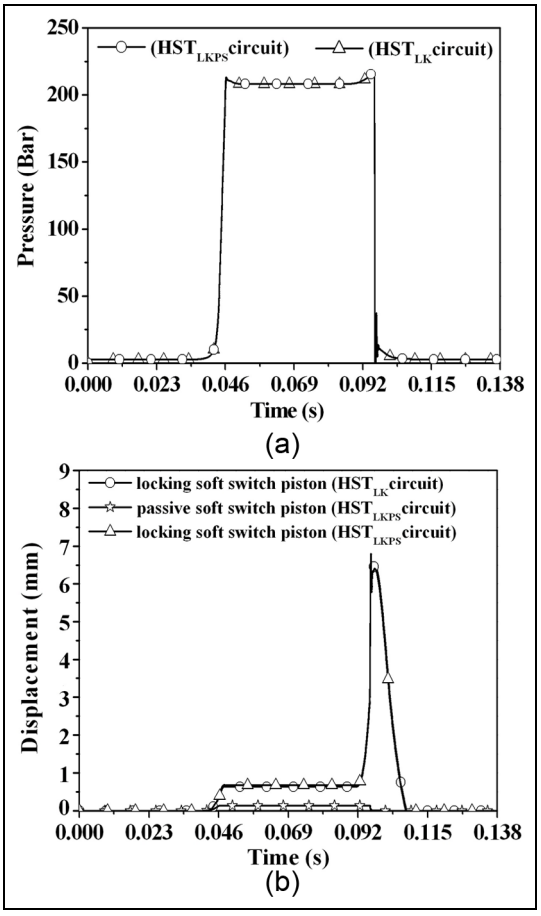

Figure 4. Comparison of the two hydraulic circuits, one with locking and passive soft switch (HST_{LKPS}) and another without passive soft switch (HST_{LK}): (a) switched volume pressure; (b) piston displacement.

locking of the soft switch. Similarly, with increase of spring preloading force, x_{crack} is decreased or vice versa. Large preload force on the spring is avoided, because it decreases the value of x_{crack}, thereby demanding more precision to manufacturing. On the other hand, lower value of preload force produces residual error during locking of the soft switch.

4.2. Comparisons of two systems: with locking and passive soft switch (HST_{LKPS}) and without passive soft switch (HST_{LK})

The time domain response of the switched volume pressure (P_{switch}) and the locking soft switch piston displacement (x_{ssl}) have been compared in Figure 4 for both systems: HST_{LKPS} and HST_{LK}. It can be observed from Figure 4 that there is no significant effect of the passive soft switching over the performance of the hydraulic system.

When the tank valve starts to close, the switched volume pressure (P_{switch}) going to increase till the load valve is fully opened, subsequently, the soft switch piston (both locking and passive) starts to travel towards the rear end of the soft switch. Note that the length of the travel of the locking soft switch piston is more than the passive soft switch piston. After a period of operation through the load valve as fully opened condition, the load valve starts to close. As a result the displacement of the locking soft switch piston is increased. Also, simultaneously the switched volume pressure (P_{switch}) is increased marginally and obtain maximum switched pressure (216 bar). At just before to fully close of the load valve, the travel length of the locking soft switch is sufficient to overcome the x_{crack} = 2.8mm length and allow the fluid flow from the rear end to the tank. At the same time, the piston gets a sudden displacement from 2.8 mm to 6.7 mm due to the spring stiffness and the pressure of the switched volume is decreased to zero.

In bond graph, the resistive element (R) is a power dissipating component, which means the power (product of effort (e) and flow (f) variables) that this element receives from the system is lost in the form of heat or some other form of energy. From Figure 3, the power loss during throttling through the DCV (load and tank valve) can be written as follows:

$$P_{throttloss} = e_5 f_5 + e_9 f_9. \tag{19}$$

The throttling energy loss can be obtained by taking time integral of the power loss, i.e.,

$$E_{throttloss} = \int P_{throttloss} dt. \tag{20}$$

The throttling energy loss (J) and maximum throttling power loss (kW) are compared in Figure 5(a) and (b), respectively, for both systems. The HST_{LK} system has been shown to exhibit marginally better performance on this yardstick. Furthermore, the total energy/power losses incurred in all the check valves and resistive elements in the soft switches along with the DCV have been computed for both the systems with an aim to investigate the influence of the passive soft switch.

Referring to the bond graph models in Figure 3, the total power losses of both systems, HST_{LKPS} and HST_{LK}, can be symbolically expressed as follows:

$$P_{totalloss} = e_5 f_5 + e_9 f_9 + e_{12} f_{12} + e_{20} f_{20} + e_{24} f_{24} + e_{27} f_{27} + e_{33} f_{33}, \tag{21}$$

$$P_{totalloss} = e_5 f_5 + e_9 f_9 + e_{12} f_{12} + e_{20} f_{20} + e_{24} f_{24} + e_{27} f_{27}. \tag{22}$$

The total energy loss is the time integral of total power loss, i.e., $\int P_{totalloss} dt$. The comparison of the total energy loss and total maximum power loss of both systems are shown in Figure 6(a) and (b).

Table 3. Comparison of power and energy losses for the two systems, HST_{LK} and HST_{LKPS}.

	HST_{LK} system	HST_{LKPS} system	Savings in HST_{LK}
Throttling power loss (maximum)	455.012 W	475.311 W	4.29%
Throttling energy loss	1.48319 J	1.533 J	3.25%
Total power loss (maximum)	8884.87 W	9129.67 W	2.68%
Total energy loss	3.66663 J	3.77792 J	2.95%

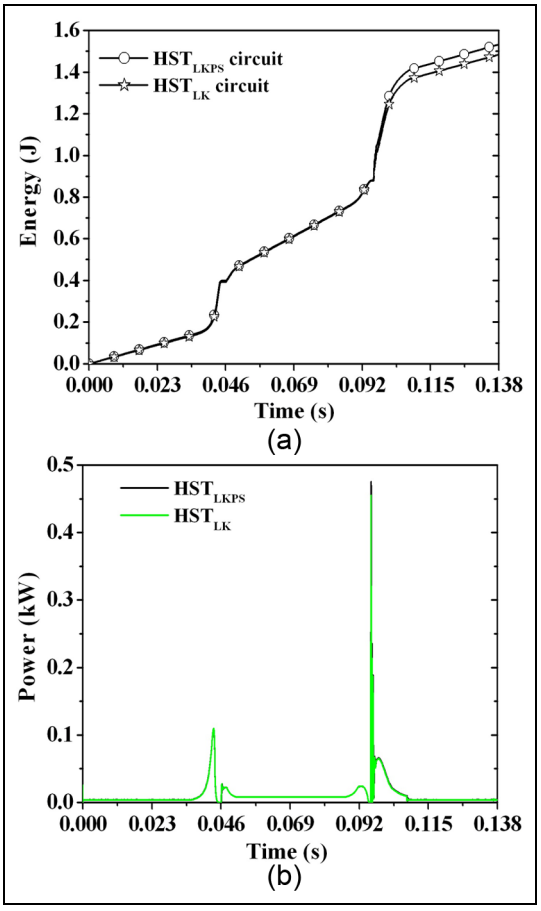

Figure 5. Comparison of the two hydraulic circuits: one with locking and passive soft switch (HST_{LKPS}) and another without passive soft switch (HST_{LK}) (a) throttling energy loss (b) throttling power loss.

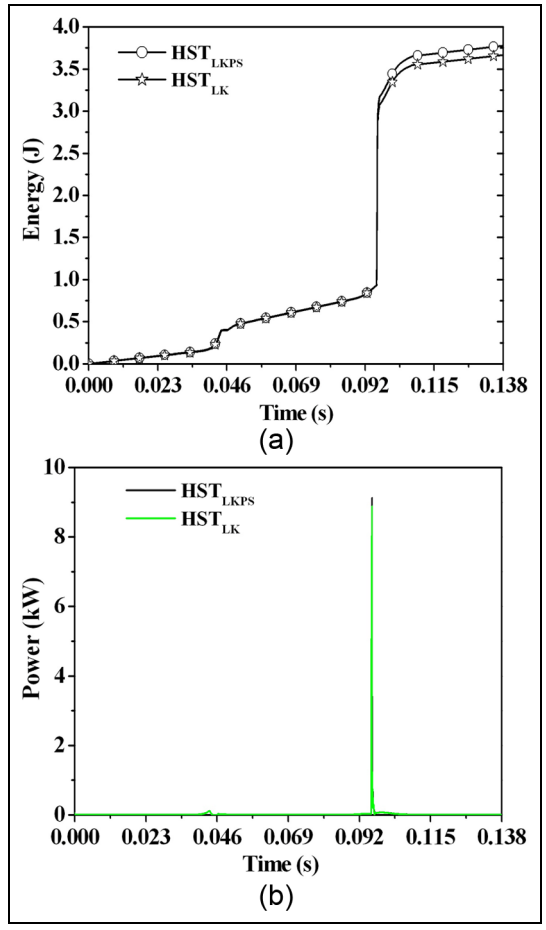

Figure 6. Comparison of the two hydraulic circuits, one with locking and passive soft switch (HST_{LKPS}) and another without passive soft switch (HST_{LK}): (a) total energy loss; (b) total power loss.

The losses for the two systems are quantified through simulation and compared in Table 3. It has been observed that the performance of both the systems is almost same. But a marginally better performance achieved in the HST_{LK} system in comparison to the HST_{LKPS} system. Also, the system can be made more compact and economical by using only the locking soft switch in the hydraulic circuit, instead of using both the locking and passive switch.

4.3. Effects of variation of different entities

The hydraulic circuit without passive soft switch (HST_{LK}) has shown to exhibit better energy efficiency as discussed in the previous subsection. Therefore, the bond graph model of the same (Figure 3(b)) is used for performance analysis as regard to the energy loss by considering variation of some critical parameters' values like valve orifice area, PWM frequency and duty cycle from their corresponding nominal values.

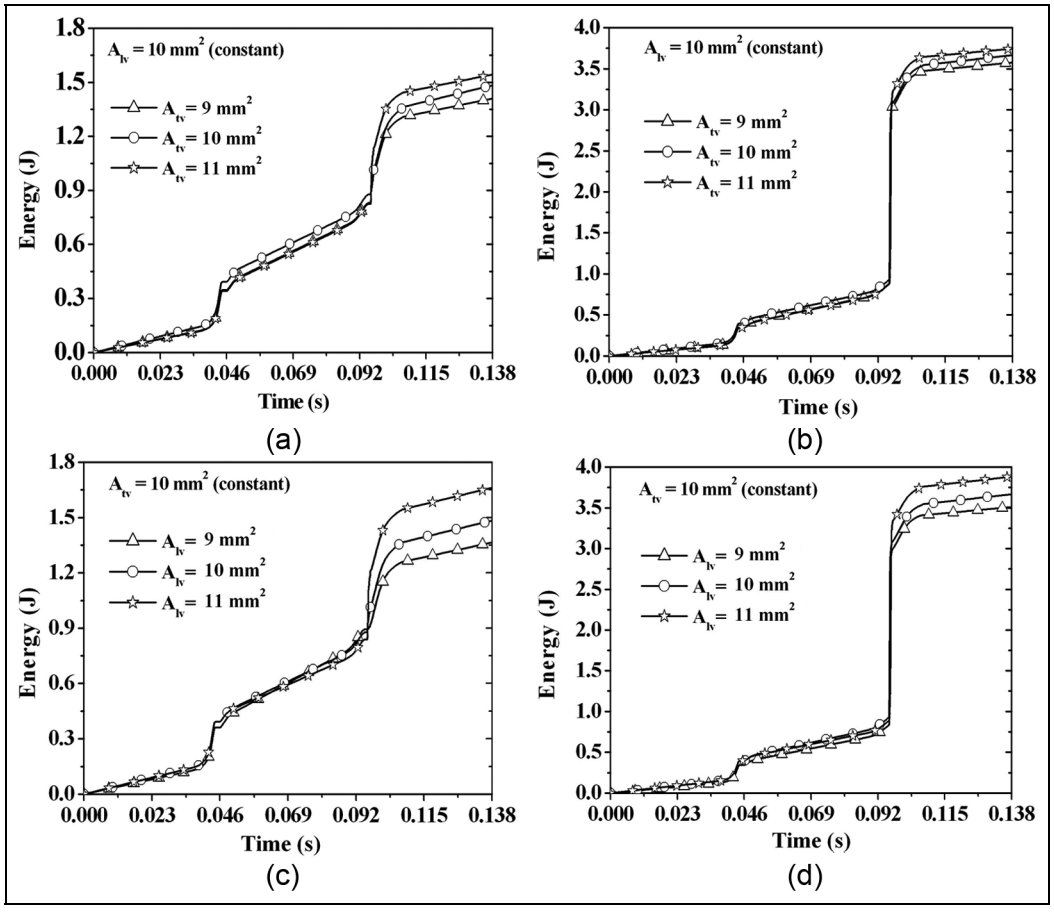

Figure 7. Comparison of energy losses on the HST_{LK} circuit for different orifice areas of the tank and load valves: (a) throttling energy losses for different A_{tv}; (b) total energy losses for different A_{tv}; (c) throttling energy losses for different A_{lv}; (d) total energy losses for different A_{lv}.

4.3.1. Influence of the orifice areas of the load and tank valves. The variation of energy losses with respect to different port opening areas of the load and tank valves are presented in Figure 7. Figure 7(a) and (b) show the variations of the throttling and the total energy losses corresponding to different tank valve orifice areas A_{tv} (i.e., 9 mm^2, 10 mm^2, and 11 mm^2) while the load valve orifice area (A_{lv}) is assigned to a constant value of 10 mm^2. Similarly, another simulation has been carried out by perturbing the load valve area A_{lv} (the different values being 9 mm^2, 10 mm^2, and 11 mm^2) for a constant tank valve area A_{tv} (10 mm^2), as shown in Figure 7(c) and (d). It can be observed that with the reduction of port opening areas of the load and tank valves, the energy losses decrease. This is due to reduced amount of fluid flow through the valves while there is a negligible increase in system pressure. Another observation that can be made comparing the Figure 7(a) and (c) is that the influence of A_{lv} (load valve area) on energy loss is more than that of A_{tv} (tank valve area). The load valve supplies fluid to the system under load set by the accumulator, while the tank valve relives the fluid to tank at atmospheric pressure or no load. Therefore, the fluid flow rate would be changed to a comparatively higher value if A_{lv} is varied resulting in significant change in throttling loss.

4.3.2. Influence of the duty cycle on energy losses. To study the effect of duty cycle, the time periods t_1, t_2, t_3, and t_4 for opening/closing of load valve and tank valve are considered to be same for simplified analysis and all of them are assigned to 0.01 s corresponding to 100% duty cycle. When the duty cycle has been changed from 100% to 80% (i.e., 0.008 s) and then 60% (i.e., 0.006 s), the corresponding variation of throttling energy losses are recorded as 1.71346 J, 1.48405 J, and 1.29114 J, respectively and total energy losses are 3.93498 J, 3.66663 J, and 3.43087 J, respectively. The variation of energy losses for different duty cycles are shown in Figure 8. As the duty cycle decreases the valve on/off time would be reduced resulting lower energy losses.

4.3.3. Influence of the PWM frequency on energy losses. Simulation studies are also made with regard to the

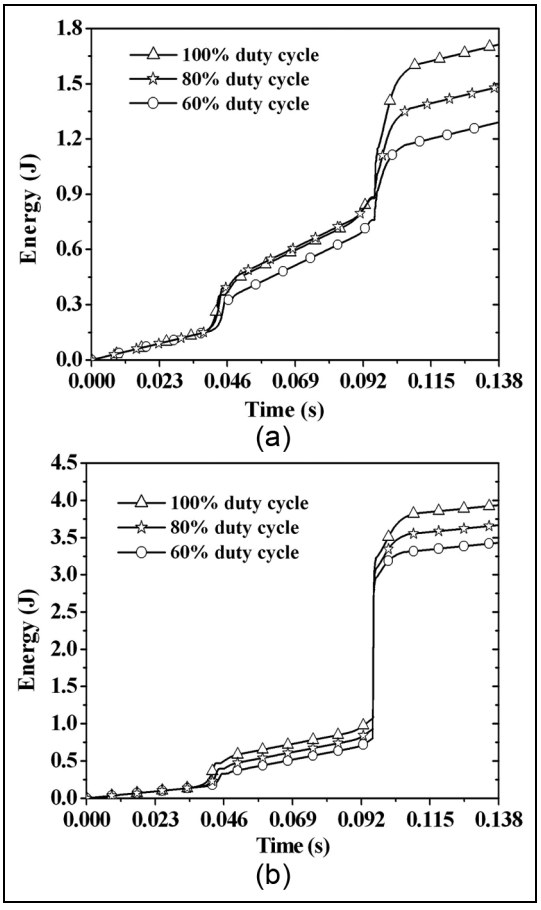

Figure 8. Comparison of energy losses on the without passive soft switch (HST_{LK}) circuit for the variation of duty cycle: (a) throttling energy loss; (b) total energy loss.

variation of the operating frequency of the load and the tank valves. In this respect, Figure 9 shows the variation of energy losses for one PWM period for the 10 Hz, 20 Hz, and 30 Hz operating frequencies of the valves. It is observed that with the increase in the PWM frequency, the throttling energy losses are decreased to 1.29081 J, 1.02834 J, and 0.88915 J, respectively, and the total energy losses are decreased to 3.47574 J, 3.21062 J, and 3.02766 J, respectively, over one period. It may be noted from Figure 9 that in the dwell period of the valve there is a constant energy loss from the system. With the increase of PWM frequency, the time period of duty cycle and the dwell time are reduced, resulting lesser amount of loss of energy over a cycle.

4.3.4. Comparative performance of the HST_{LKPS} and HST_{LK} for variations of nominal parameters. The comparative performance analysis of HST_{LK} over HST_{LKPS} is shown in Table 4. This study is based on the model simulation with the variations in parameters A_{tv}, A_{lv}, duty cycles and PWM frequencies and the energy loss calculation over one cycle of operation of both systems.

In practice, valve areas are not changed too much because they govern the flow rate and consequently the load carrying capacity. Likewise, duty cycle and PWM frequency variations relate to speed of operation and there are also limits on those because too slow operation reduces productivity and too fast operation creates a lot of dynamic (inertial) load. So the ranges of variations used here are kept within practical limits.

5. Conclusions

This study confirms that the HST_{LK} system is better as compared to the HST_{LKPS} system so far as energy savings are concerned. Although the simulation results for both the hydraulic systems are not much different, the HST_{LK} system is more efficient, simple and compact. Also, this system is economical because it saves the manufacturing cost for the passive soft switch. The locking soft switch parameters have been optimized (through multi-run bond graph simulation) with respect to the minimum throttling energy loss. During optimization, large spring stiffness and large preload on the spring have been avoided because this would yield a lower value of x_{crack}, which is impossible to fabricate. Much lower values of the same parameters have

Table 4. Comparative performance of the HST_{LKPS} and HST_{LK}.

Varying parameters (others fixed)	Nominal value	Modified value	Additional % energy saving in HST_{LK} over HST_{LKPS}
A_{tv}	10 mm^2	9 mm^2	+6.5%
		10 mm^2	+4.29%
		11 mm^2	+3.14%
A_{lv}	10 mm^2	9 mm^2	+6.8%
		10 mm^2	+4.29%
		11 mm^2	+3.1%
Duty cycle	100%	100%	+3.77%
		80%	+3.25%
		60%	+2.69%
PWM frequency	20 Hz	10 Hz	+9.09%
		20 Hz	+8.6%
		30 Hz	+6.5%

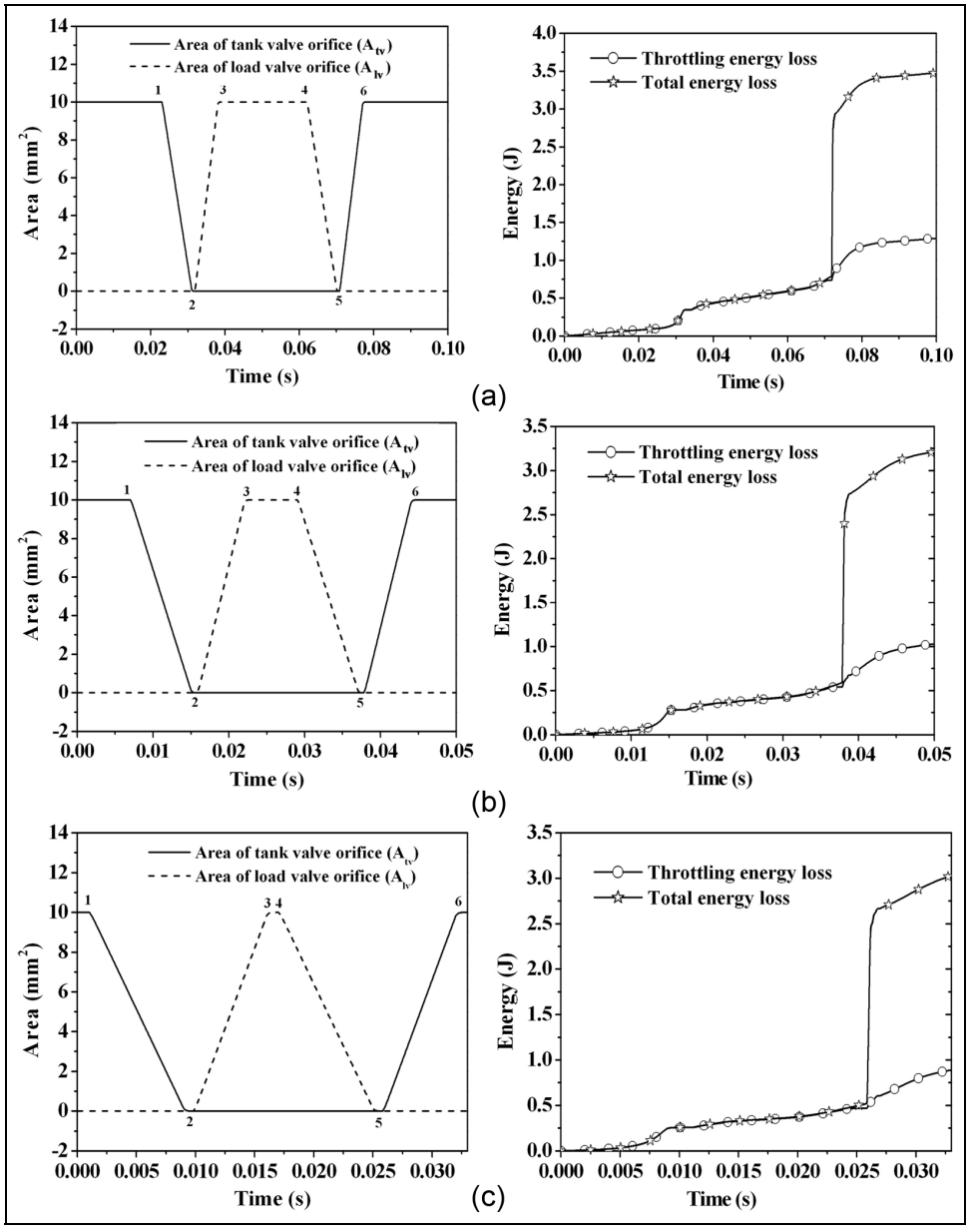

Figure 9. Comparison of energy losses for different PWM frequencies.

also been avoided, because these would produce residual errors during locking and unlocking of the soft switch. Finally, bond graph simulation of the HST_{LK} system has been carried out for performance analysis. A study on performance variation due to variation of some critical parameters like valve orifice areas, duty cycles and PWM frequencies is performed and it is shown that smaller valve discharge leads to less throttling losses, reduced dwell period (more duty cycle) increases energy efficiency and more PWM frequency leads to more losses.

This simulation study is based on a simple pedagogical hydraulic circuit. In future, this locking type soft switching concept would be applied to a more complex power hydraulic system for experimental corroboration. Furthermore, if the high pressure line would be connected with a speed reversal circuit, it would alternatively change from high pressure to low pressure. In such a case, the check valve (used for system's safety) outflow connection from the rear end of the locking soft switch to the pressure line should be altered to obtain the full benefit of regeneration. An appropriate design for that may be taken up in future.

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